GGG-D-780a

February 18, 1965

SUPERSEDING
Int. Fed. Spec. GGG-D-00780(GSA-FSS)
August 11, 1964

FEDERAL SPECIFICATION

DRIVER, PROJECTILE UNIT, POWDER ACTUATED (TRIGGER-FIRED, LOW VELOCITY, PISTON TYPE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers the requirements for one type of low velocity powder actuated driver used to drive projectiles (drive pins) into dense material such as, steel or concrete. For above water use only.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

FF-P-395—Pin, Drive, Guided; Pin, Drive, Powder Actuated.

QQ-S-741—Steel, Carbon: Structural Shapes, Plates, and Bars.

PPP-B-585—Boxes, Wood, Wirebound.

PPP-B-591—Boxes, Fiberboard, Wood-Cleated.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-T-60—Tape; Pressure Sensitive Adhesive, Waterproof—for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards

FSC 5130

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–P–116—Preservation, Methods of.

MIL–B–121—Barrier Material, Greaseproofed, Flexible (Waterproofed).

MIL–P–3542—Primer; Pressure-Sensitive Tape.

MIL–L–10547—Liners, Case, Waterproof.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Governmental:

National Bureau of Standards (NBS) Handbook:

H28—Screw-Thread Standards for Federal Services.

(Copies may be obtained upon application, accompanied by check, money order, coupon, or cash, to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402. Price may be obtained from the superintendent of Documents.)

3. REQUIREMENTS

3.1 Illustration. The illustration shown herein is for the convenience of identification, and is not intended to preclude the purchase of drivers that are otherwise in accordance with the requirements of this specification.

3.2 Materials. The materials used in the construction of the drivers shall be new and shall be suitable for the purpose intended.

3.3 Construction. Drivers shall be rugged in construction and reliable in operation. All parts of the driver shall be mounted, secured, and reinforced to withstand external abuses to which they may be subjected during normal usage and also, the internal stresses imposed by the strongest commercially available powder load which will chamber in the driver. All drivers shall conform to the requirements of this specification and shall be capable of withstanding the tests specified herein.

3.4 Design. Drivers shall be similar to figure 1 and consist of a barrel, piston, trigger actuated firing mechanism, and handle. Drivers shall be designed to utilize the explosive force of a powder charge (cartridge) to drive pins into dense material such as, steel or concrete. The design shall utilize a limited stroke piston between the cartridge and the drive pin to transfer the energy from the cartridge to the drive pin.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

The driver shall be capable of efficient operation in any position. The driver shall be so designed that the cartridge and drive pin can be loaded and unloaded easily and quickly. Means shall be provided to hold the drive pin in the driver while the cartridge is loaded and the driver is brought into position for firing.

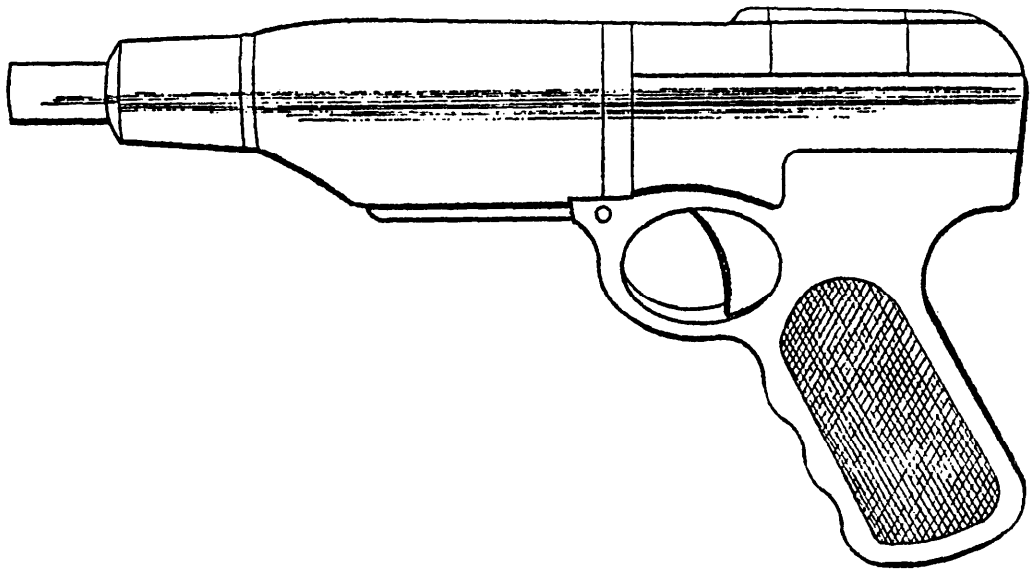


FIGURE 1. Driver, projectile unit, powder actuated

3.4.1 *Threads.* Threads shall conform to the applicable requirements of NBS Handbook H28.

3.4.2 *Finish.* All drivers shall be finished smooth and free of burrs and sharp projections.

3.4.3 *Drive pin capability.* The driver shall be capable of properly driving all drive pins that are in accordance with types I (except type I, size number 2) and II of FF–P–395.

3.4.4 *Cartridges and cartridge container.* Cartridges used to actuate the driver shall be .22 caliber, rimfire, crimped (without wads), charged with smokeless powder, and primed with a noncorrosive mixture. The powder charge within each cartridge shall be identified by the following colors.

Powder charge	Color
1 (Sub-light)	Gray
2 (Extra light)	Brown
3 (Light)	Green
4 (Medium)	Yellow

Each cartridge and cartridge container shall be permanently marked with the proper identifying color.

3.5 **Safety.** All drivers shall be reliable in operation for safety in use. The firing mechanism shall be so designed that the driver cannot fire during loading or prepa-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

ration to fire; or if the driver is dropped while loaded (see 4.5.1). Firing of the driver shall be dependent upon at least two separate and distinct operations. The final firing movement shall be separate from the operation of bringing the driver into the firing position. All drivers shall be so designed that they cannot be fired unless the muzzle end is held against the work surface with a force of at least five pounds greater than the total weight of the driver.

3.5.1 *Velocity.* The velocity of the drive pin shall be not greater than 300 feet per second when measured 6.5 feet from the muzzle end of the barrel when tested in accordance with 4.5.2.

3.6 Component parts.

3.6.1 *Barrel and piston.* The barrel and piston shall be of high-grade alloy steel, heat-treated to develop sufficient strength to resist the stress imposed by the shock of the exploding cartridge and resist permanent deformation in the event the barrel becomes accidentally plugged or if the piston bottoms in the barrel under maximum driving force with no resistance from the projectile.

3.6.2 *Firing mechanism.* The firing mechanism shall be of heat-treated, high-grade steel, properly protected against corrosion. The firing pin shall be locked in position away from the cartridge to prevent accidental firing. The firing mechanism shall be so designed that repeated firing or dirt accumulation in normal use will not cause malfunction.

3.6.3 *Handle.* The handle shall be properly proportioned and of suitable shape and size to provide a comfortable nonslip handgrip.

3.7 Operator's instruction and maintenance manual. A manual shall be supplied with each driver furnished. The manual shall include instructions for a step-by-step proper operating procedure and clearly illustrate all parts necessary in the maintenance of the driver. The instruction shall include the proper methods of cleaning, oiling and care of the driver to assure satisfactory performance. The manual shall also contain the following mandatory safety regulations:

AUTHORIZED PERSONNEL

(a) Personnel operating drivers shall be instructed and certified by the manufacturer's representative, or a responsible official of the using activity, in the operation of each tool.

(b) In the course of instructions, safe practices in the use of the drivers shall be stressed and the personnel certified to operate the tool shall be cautioned that unqualified persons shall not be allowed to operate or handle the tools at any time.

IDENTIFICATION CARDS

(a) Personnel certified to operate drivers shall be issued identification cards signed by a responsible official of the using activity.

(b) Identification cards shall clearly state the manufacturer's name and the class and style of tool for which the operator has been instructed and certified to use.

(c) A master record of identification cards issued shall be maintained at the point of issue. This record shall correspond with the identification cards issued to the operators.

(d) In no instance shall a tool be issued to an operator other than the tool of the manufacturer's class and style shown on the identification card issued to the operator.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.

STORAGE AREA

(a) When not in use, the driver, cartridges, and drive pins shall be stored in a waterproof area under lock and key.

(b) Only authorized personnel shall have access to the storage area.

(c) Special precautions shall be taken to prevent unauthorized personnel from having access to the storage area.

MATERIALS AND DRIVING DISTANCES

(a) The manual shall include a list of materials unsuitable for driving fasteners.

(b) Materials on which the driver is to be used shall be such as to offer enough resistance to prevent the fastener from passing completely through the material.

(c) Fasteners shall be driven not less than the following distance from the edge of a structural surface:

2 inches minimum for masonry materials.

½ inch minimum for metals.

(d) Materials capable of stopping the fastener shall be placed directly behind the work area, if necessary to insure safety to the personnel.

(e) Preliminary tests of the strength of the material shall be made before starting the work operations.

OPERATING PRECAUTIONS

(a) Personnel certified to operate drivers shall be provided with chipper goggles or face shield, and a hard hat.

(b) The driver shall never remain loaded when not in use.

(c) The driver shall never be loaded while being transported.

(d) If for any reason the work is interrupted after loading, the driver shall be unloaded immediately.

(e) In case of misfire, the driver shall be held in the operating position for not less than 30 seconds. The driver shall be opened and the cartridge shall be removed before removing the driver from the working surface.

(f) At no time shall an attempt be made to experiment or make adjustment to the driver, except by instructions from the manufacturer.

(g) The driver shall not be used where there is evidence of a combustible atmosphere.

(h) Cartridges and drive pins shall be kept in their respective containers and separate from other equipment.

(i) Defective cartridges shall not be thrown into trash containers or otherwise discarded. They shall be returned and properly disposed of by the issuing personnel.

(j) Cartridge shall be kept in a metal container with the word "EXPLOSIVE" conspicuously lettered thereon.

(k) The operator shall begin with light strength cartridges and progress to the stronger strength

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

cartridges when determining the correct cartridges for a particular application.

CONTACT WITH MANUFACTURER

(a) The manufacturer shall be contacted immediately if any difficulty is experienced in the use of the driver.

(b) Defective drivers shall be removed from service until proper repairs have been completed.

3.8 Marking. Each driver shall be marked in a plain and permanent manner or shall be provided with a suitable nameplate securely attached to the tool. The marking shall designate the manufacturer's name or a trademark of such known character that the source of manufacture may be readily determined. The words "USE ONLY IF CERTIFIED OPERATOR" shall also be included in the marking.

3.9 Repair parts. The successful bidders supplying drivers covered by this specification shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable, replaced or adjusted without requiring modification.

3.10 Carrying case. A carrying case shall be provided for each driver. The case shall be of sheet steel, fiberglass or plastic and shall be substantial, durable, and of rigid construction suitable for storage and transportation. The case shall be provided with a partitioned section or a removable tray for carrying cartridge, fasteners, tools, and other equipment required for operation of the driver. The case shall be provided with a substantial carrying handle and with either a lockable catch or hasp with staple for retaining and locking the hinged cover in a closed position. The interior and exterior of steel cases shall have a durable coating of paint or enamel.

3.11 Cleaning and maintenance tools. Cleaning and maintenance tools normally furnished in commercial practice, shall be provided with each driver.

3.12 Workmanship. Workmanship shall be first class throughout and free from any imperfections which would impair the serviceability or create hazardous conditions under normal operating conditions.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* All drivers manufactured by a single production process and offered for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.2.2 *Sampling for examination.* A random sample shall be selected from each lot for examination in accordance with MIL–STD–105 at inspection level II. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.2.3 *Sampling for performance tests.* A random sample shall be selected from each lot for performance tests in accordance with

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

MIL–STD–105 at inspection level II. The AQL shall be 1.5 percent defective.

4.3 Examination. Each driver selected in accordance with 4.2.2 shall be visually examined to determine compliance with this specification. Any driver in the sample containing one or more defects shall be rejected.

4.4 Performance test. Each driver selected in accordance with 4.2.3 shall be tested for performance by driving three pins 3-inches long through a standard two-inch piece of lumber and into a ¼-inch steel plate and three pins ¾-inch long through a ¼-inch steel plate. The pins shall not be started by hand or driven through holes. The pins shall conform to type I of FF–P–395. The steel plate shall conform to grade A of QQ–S–741 with a Brinell hardness of 126 to 140. The cartridge used shall be of sufficient strength to seat the pin. The driver shall be rejected if any malfunctions occur.

4.5 Drop and velocity tests. The manufacturer shall certify that each driver being offered is of the same material, design, construction, and dimensions as a driver that has previously passed the following drop and velocity tests.

4.5.1 *Drop test.* The driver shall be loaded with a live cartridge and dropped three times, from a height of ten feet, onto a concrete surface, in such a way that it will land on the muzzle, on the handle, and on the side. The driver shall be rejected if it fires as a result of the test or if it does not meet the requirements of 3.5 after the test.

4.5.2 *Velocity test.* The driver shall be tested to determine the velocity of a drive pin having a maximum weight of 0.072 ounces at a distance of 6.5 feet from the muzzle end. The test shall be conducted in accordance with accepted ballistic test methods, using a medium (yellow) strength cartridge. The driver shall be rejected if the velocity of the drive pin exceeds 300 feet per second.

4.6 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking requirements as required by section 5 of this specification are complied with. Defects shall be scored in accordance with table I. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S–1 and the AQL shall be 6.5 defects per hundred units.

TABLE I. *Classification of preparation for delivery defects*

Examine	Defect
Markings (exterior and interior containers.	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as; incomplete closure of container, loose strapping. Bulging or distortion of containers.
Packaging and packing	Packages improperly sealed: Tape not as specified, joints not waterproof sealed. Contents more or less than specified.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 *Level A.* Component parts of the driver subject to corrosion shall be cleaned by process C–1 and coated with type P–9 preservative in accordance with MIL–P–116. The driver shall be wrapped with greaseproof barrier material conforming to MIL–B–121, type I, grade A class 2, class 2, secured with pressure sensitive tape conforming to PPP–T–60, type III, class 3, and unit packaging in accordance with method 1A–2 or 1A–14 of Military Specification MIL–P–116, using a fiberboard box conforming to PPP–B–636.

5.1.1.1 *Intermediate packaging.* The driver and publications comprising a complete unit shall be packaged together in the carrying case. Cushioning shall be provided within the case. The lid shall be closed and secured with the means provided. The lid joints shall be sealed with pressure-sensitive tape conforming to PPP–T–60, type III, class 1. Surface to which the tape will adhere shall be coated with pressure- sensitive tape primer conforming to MIL–P–3542, before the tape is applied.

5.1.2 *Level C.* The driver and carrying case shall be preserved and packaged to afford protection against deterioration and damage during shipment. Packaged quantities shall be as specified.

5.2 Packing. Packing shall be A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* Intermediate packages of like description shall be packed in quantities not to exceed 200 pounds gross weight in overseas type snug-fitting nailed wood boxes conforming to PPP–B–621 (class 2), or wood cleated plywood boxes conforming to PPP–B–601. The boxes shall be closed and strapped in conformance with the appendix to the applicable box specification. Shipping containers shall have case liners conforming to MIL–L–10547 and shall be closed and sealed in accordance with the appendix thereto.

5.2.2 *Level B.* Intermediate packages of like description shall be packed in domestic type snug-fitting wood cleated fiberboard, wood cleated plywood, nailed wood, or wirebound wood boxes conforming to PPP–B–591, PPP–B–601, PPP–B–621, and PPP–B–585, respectively. Gross weight shall not exceed 200 pounds.

5.2.3 *Level C.* The driver and carrying case shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military activities.* In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Level of packing and packaging required (see 5.1 and 5.2).

(*c*) Any special marking required (see 5.3).

6.2 Accessories. Most drivers are so designed that shields, guards, jigs or fixtures

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.

can be attached. These accessories are available from driver manufacturers. They are not generally interchangable between different makes of drivers and should be used only with the make of driver and the application for which they are designed.

6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Tools, not otherwise indexed by name.

Carload minimum weight 30,000 pounds.

Motor:

Tools, hand, not otherwise indexed.

Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

DOD coordination has been waived (No interest).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.